



The Effects of Set Structure Manipulation on Chronic Adaptations to Resistance Training: A Systematic Review and Meta-Analysis

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Abstract

Background The acute effects of resistance training (RT) set structure alteration are well established; however, less is known about their effects on chronic training adaptations.

Objective The aim of this systematic review and meta-analysis was to synthesise the available evidence on the effectiveness of traditional (TS), cluster (CS) and rest redistribution (RR) set structures in promoting chronic RT adaptations, and provide an overview of the factors which might differentially influence the magnitude of specific training adaptations between set structure types.

Methods This review was performed using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines encompassing the literature search of five databases. Studies in English that compared muscular strength, endurance, and/or hypertrophy adaptations, as well as vertical jump performance, velocity and power at submaximal loads and shifts in the slopes of force–velocity profiles between TS and CS or RR set structures (i.e., alternative set structures) were included. Risk of bias assessment was performed using a modified Cochrane Collaboration’s tool for assessing risk of bias in randomised trials. Random-effects meta-analyses and meta-regressions were performed where possible.

Results 17 studies met the inclusion criteria, none had more than one risk of bias item assessed as high risk. Pooled results revealed that none of the set structures were more effective at inducing strength (standardised mean difference (SMD) = − 0.06) or hypertrophy (SMD = − 0.03). TS were more effective at improving muscular endurance compared to alternative set structures (SMD = − 0.38), whereas alternative set structures tended to be more effective for vertical jump performance gains (SMD = 0.13), but this effect was not statistically significant ($p = 0.190$). Greater velocity and power outputs at submaximal loads (SMD = 0.18) were observed when using alternative set structures compared to TS. In addition, alternative set structures promoted greater shifts of the slope of force–velocity profiles towards more velocity dominant profiles compared to TS (SMD = 0.28). Sub-group analyses controlling for each alternative set structure independently showed mixed results likely caused by the relatively small number of studies available for some outcomes.

Conclusion Modifying TS to an alternative set structure (CS or RR) has a negligible impact on strength and hypertrophy. Using CS and RR can lead to greater vertical jump performance, velocity and power at submaximal loads and shifts to more velocity dominant force–velocity profiles compared to training using TS. However, TS may provide more favourable effects on muscle endurance when compared to CS and RR. These findings demonstrate that altering TS to alternative set structures may influence the magnitude of specific muscular adaptations indicating set structure manipulation is an important consideration for RT program design.

Protocol registration The original protocol was prospectively registered (CRD42019138954) with the PROSPERO (International Prospective Register of Systematic Reviews).

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Key Points

Set structure manipulation during resistance training has a negligible impact on muscular strength gains and hypertrophic outcomes.

While traditional set structures are preferable to alternative set structures (i.e., cluster and rest redistribution set structures) when the goal is to improve muscular endurance, alternative set structures tend to be more effective at improving vertical jump performance compared to traditional set structures.

Greater velocity and power outputs at submaximal loads and consequent shifts of the slope of force–velocity profiles towards a more velocity dominant profile can be expected after resistance training using alternative set structures compared to traditional set structures.

1 Introduction

Resistance training (RT) is considered to be the principal mode of exercise for improving muscular strength, power and hypertrophy. While these adaptations often coincide with improvements in jumping, sprinting, agility, and sport-specific performance [1, 2], improvements in muscular strength, power, and muscle mass may also decrease the risk of falling in the elderly and might positively affect cognitive function, psychosocial status, and the quality of life [3–7]. However, the magnitude of their improvements is influenced by the structure of the training program [8]. For instance, training volume, intensity, frequency, rest intervals, exercise selection, exercise order, proximity to muscular failure, repetition velocity, and set structure can all affect the resultant training adaptations. As such, there has been a great deal of research into the programming and manipulation of RT variables. While the influence of most of these variables has been extensively reported in the literature and summarised through recent systematic reviews and meta-analyses [9–14], the effect of set structure manipulation on muscular strength, power and hypertrophy is comparatively neglected [15].

Traditionally, the completion of a training set occurs without any rest being taken between repetitions. Subsequently, this basic set configuration, often referred to as a traditional set structure (TS), is continued for a predetermined number of sets prescribed in a training session. In addition, sport professionals often instruct athletes to perform concentric muscle actions as quickly as possible since explosive concentric muscle actions often result in greater

training effects compared with intentionally slower concentric actions [16–18]. However, when using TS, concentric velocity decreases due to fatigue as the number of repetitions increases [19–22]—this velocity loss is observed across a variety of simple and complex RT exercises and is aggravated by multi-set training with a high number of repetitions. Fatigue-induced velocity loss and concomitant decreases in power outputs result in an increased overall time under tension [23, 24] and increased myoelectrical activity toward the end of a set [25–27], both of which might be beneficial for inducing strength and hypertrophy adaptations [28–31]. This has led some practitioners and researchers to believe that TS structures seem ideal for promoting both muscular strength and hypertrophy [30, 32–34]. While fatigue is often thought to be of paramount importance for the development of muscle mass and strength, more recent literature [10, 12, 14, 35–37] repeatedly demonstrates that training volume likely plays a key role in both strength and hypertrophic adaptations as well. In addition, a substantial body of research now indicates that performing repetitions at maximal intended velocity [16–18] and terminating training sets once a moderate velocity loss is achieved during a training set [21, 38–40], may be more important for inducing maximal strength gains than increasing time under tension. This, coupled with the widely acknowledged deleterious effects of fatiguing exercise on muscular power development and performance of athletic tasks [21, 38, 41, 42], highlights a need for strategies that limit the typical fatigue-induced reductions in movement velocity often seen during TS. Therefore, alternative set structures, which consist of shorter but more frequent rest periods, have gained considerable attention from the sport science community for the purpose of mitigating acute fatigue during RT.

The simplest and most effective way to off-set acute fatigue and maintain high-movement velocities is to add pre-planned rest intervals within a set alongside normal inter-set rest periods, thereby breaking sets into small clusters of repetitions [15, 43]. Although these so-called “cluster sets” (CS) are frequently used to describe any set structure in which rest periods are more frequent than TS (see Tufano et al. [15] for more detailed discussion), CS simply represent a strategy of adding short rest periods within a set while maintaining normal rest periods between sets to maximise performance of each individual repetition within sets [15]. Thus, an indispensable feature of CS is that they extend total training time relative to TS [44–46]. From a practical standpoint, this might not always be feasible for sport professionals, especially for those working with athletes who train multiple times a day and who need to manage RT fatigue for both game performance and sport-specific training sessions. To overcome these constraints, researchers have investigated other possible alternatives to CS, such as simply redistributing the total rest time of TS structures, to include shorter

and more frequent rest intervals. While this set structure, known as rest redistribution (RR), can effectively alleviate acute mechanical [47–49], metabolic [50, 51] and perceptual [52, 53] markers of fatigue during RT, it does so without increasing total training time associated with CS. Indeed, Jukic et al. [54] recently reported alternative set structures were generally superior to TS in a systematic review and meta-analysis on their effects on mechanical, metabolic and perceptual responses to RT. However, conclusions from this review were limited to the acute effects of alternative set structures on RT performance and fatigue management; the generalisability of the findings to chronic adaptations, such as muscular strength, power and hypertrophy, is uncertain.

Considerable evidence is accumulating from longitudinal studies (> 4 weeks in duration) comparing the effectiveness of TS and alternative set structures on various training adaptations. However, the findings appear equivocal. For instance, Goto et al. [34], Drinkwater et al. [32] and Lawton et al. [33] all emphasised the importance of fatigue during RT in inducing strength gains, as they observed greater increases in strength in groups using TS rather than alternative set structures. In contrast, Oliver et al. [55] suggested that shorter, but more frequent rest periods allow for greater improvements in both strength and power, while Hansen et al. [42] concluded that CS present a viable option for the development of lower-body power, but not strength, when compared to TS. To further complicate matters, other studies reported no significant differences in muscular strength and power-related outcomes when comparing TS to CS or RR [56–61].

Therefore, these inconsistencies in the literature regarding the long-term effects of alternative set structures on training adaptations justify the need for a comprehensive review and synthesis of the available evidence. Thus, the aim of this systematic review and meta-analysis was to synthesise the available evidence on the effectiveness of TS, CS and RR structures in promoting chronic RT adaptations. To achieve further clarity on the topic, this review also aimed to provide an overview of the factors which might differentially influence the magnitude of specific muscular adaptations between the set structure types, allowing sport professionals to compare and contrast them with more nuance. Such evidence is important to guide RT prescription and ultimately determine the most effective set structure to promote an intended training adaptation.

2 Methods

2.1 Registration of Systematic Review Protocol

A systematic review of the literature was performed according to the guidelines in the *Cochrane Handbook for*

Systematic Reviews of Interventions (version 5.1.0) and following the checklist for the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [62]. The original protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO) in June 2019 and was part of a larger systematic review project comparing the acute and chronic effects of alternative set structures across a range of outcomes (Registration number: CRD42019138954). The protocol registration occurred before any of the searches were conducted.

2.2 Literature Search

A patient/population, intervention, comparison and outcomes (PICO) strategy which consisted of terms for traditional, cluster and rest redistribution set structures; resistance training and mechanical, perceptual, and metabolic outcomes as well as muscular strength, power and hypertrophy adaptations was used to build search criteria for electronic databases. The following bibliographic databases were searched from inception to June 6th, 2019: PubMed/MEDLINE, SCOPUS, CINAHL (Cumulative Index to Nursing and Allied Health), SPORTDiscus and Web of Science. The search syntax adapted for each database and applied to the title, abstract, and keyword search fields is reported in the Supplementary Information Appendix S1. Secondary searches included: (a) screening the reference lists of all included studies and relevant review papers; (b) examining the studies that cited the included studies (i.e., forward citation tracking) through Google Scholar; (c) search alerts to monitor any new search results after the date of the last search up to June 3, 2020. Free-text terms were chosen based on pilot searches that achieved an appropriate balance between search sensitivity and precision without the use of controlled vocabulary (e.g. MeSH [Medical Subject Headings]). No year restrictions were applied in the search. The strategy included only terms relating to or describing the intervention. All trials had to be peer-reviewed and published in an English-language journal while conference abstracts and proceedings were not considered. Three authors (IJ, AGR, and BVH) independently performed the searches. Records were downloaded into Endnote (version X8.2, Clarivate Analytics, Philadelphia, PA, USA) and duplicates were removed with automated and manual methods before being screened by title and abstract. The full texts of the remaining records were then retrieved and assessed for eligibility. Any discrepancies between the authors in the study selection were resolved in consultation with a fourth reviewer (JJT).

2.3 Eligibility Criteria

All studies included in this systematic review met the following inclusion criteria: (1) the study was published in English; (2) the study compared muscular strength, power-related, endurance, and/or hypertrophy adaptations between TS and CS or RR set structures; (3) the study included an intervention group (CS) during which additional intra-set rest intervals were included and total rest periods extended compared to the control group, and/or a second intervention group (RR) during which total rest periods were the same compared to control group; (4) the study included a control group (TS) in which multiple repetitions within a set were performed in a consecutive fashion while including only rest periods between the sets (i.e., inter-set rest); (5) training volume (set $\times$ repetitions) and training frequency were matched between the conditions; (6) in studies that assessed muscular strength and hypertrophy adaptations, both lean body mass changes and changes at the muscle level (whole muscle or muscle fiber) were considered regardless of assessment method (e.g. ultrasound, dual-energy x-ray absorptiometry (DEXA) or tape with skinfolds), while for muscle strength, tests with a repetition maximum (RM) component (e.g. 1RM, 5RM), or maximum voluntary contraction (MVC) tests were considered; (7) in studies that investigated expressions of muscular power, such as changes in jump height, sprint and change of direction times, as well as velocity and power outputs at submaximal loads were considered; (8) the interventions lasted a minimum of 4 weeks; (9) participants had no known medical conditions or injuries.

2.4 Study Coding and Data Extraction

From the studies that met the inclusion criteria, the following data were extracted into an Excel spreadsheet: (1) study identification information; (2) adherence and study duration; (3) sample size; (4) participants' characteristics, including age, body mass, body height, sex, strength levels, and training experience; (5) resistance training prescription details for TS, CS, and RR set structures; and (6) means and standard deviations as well as raw mean changes and standard deviations of changes for pre- and post-intervention assessments of the relevant outcome measures. If insufficient data were reported, the authors of those studies were contacted by-email. Web Plot Digitizer software (Version 4.1; TX, USA) was used to extract data from figures when the authors did not report or provide the data. Data extraction was completed independently by two authors (IJ and BVH) using a pilot-tested form on five randomly selected studies which was then modified accordingly. Coding files were cross-checked between the authors, and any observed differences were resolved via discussion and agreement, or with a third reviewer (JJT).

2.5 Risk of Bias Assessment

Risk of bias assessment was performed using a modified Cochrane Collaboration's tool for assessing risk of bias in randomised trials [63]. Modifications included removal of the performance bias criterion and adding set structure bias, outcome assessment bias, and familiarisation bias. In addition, while the blinding of outcome assessment bias was not excluded, it was considered of trivial importance in the present review since the training volume and intensity of load were matched between the interventions. Therefore, we did not expect that awareness of intervention allocation would introduce bias. Similar to previous systematic reviews and meta-analyses of exercise intervention studies [13, 64], the performance bias criterion was removed because it was considered impossible to successfully blind participants and personnel in supervised exercise intervention studies. Set structure bias was considered to be important for this review and was related to whether all exercises during the intervention were performed using a single-set structure. For instance, if the experimental group performed only one or two exercises using CS, but performed the remaining exercises using TS, this may have introduced bias. Outcome assessment bias was also considered important and was related to the reliability and appropriateness of the equipment or assessor that were used to assess the outcomes of interest. For instance, if previous research showed the reliability of a given device used to record kinetic or kinematic variables to be poor, this may have induced bias (e.g. using a linear position transducer whose reliability was not previously established to assess changes in movement velocity at submaximal loads). Familiarisation bias was related to whether the participants were adequately familiarised with the protocols in a given study. If the authors did not clearly state that a familiarisation session took place or that their participants were regularly performing exercises and performance tasks used in a given study, bias may have risen from the participants being unaccustomed to the study protocols. Risk of bias was assessed based on the information reported in the published paper and not on information provided by authors (i.e., Tables 1 and 2 of this review). Assessments were completed independently by two reviewers (IJ and AGR) while any observed differences were resolved via discussion and agreement before merging the scores into a single spreadsheet. Risk of bias was considered in the interpretation of the results by applying the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system, similarly to previous reviews investigating physiological and biomechanical outcomes between different exercise modalities [65]. Specifically, the overall quality of the evidence synthesis was rated as

Table 1 Characteristics of individual study samples

Study	Participants	Sex: M/F	Age (years)	Height (cm)	Mass (kg)	Training experience (subjective description; years of experience)
Asadi et al. (2016) [58]	TS group: n = 7	Unclear	20 ± 1	180 ± 5	78 ± 4	Physically active sport science students; unclear
	RR group: n = 6	Unclear	21 ± 1	180 ± 3	79 ± 3	Physically active sport science students; unclear
Carneiro et al. (2019) [59]	TS group: n = 8	0/8	63 ± 7 ^a	160 ± 1 ^a	70 ± 17 ^a	Healthy postmenopausal women; unclear
	CS group: n = 8	0/8				Healthy postmenopausal women; unclear
Davies et al. (2019) [60]	TS group: n = 10	5/5	25 ± 7	Unclear	76 ± 10	Resistance-trained individuals; 5 ± 8
	CS group: n = 11	7/4	26 ± 7	Unclear	74 ± 10	Resistance-trained individuals; 4 ± 4
Dias et al. (2020) [79]	TS group: n = 35	0/35	61 ± 5 ^a	156 ± 5 ^a	65 ± 13 ^a	Physically active elderly women; unclear
	CS group: n = 31	0/31				Physically active elderly women; unclear
Farinas et al. (2019) [80]	TS group: n = 12	8/4	24 ± 5	173 ± 8	71 ± 11	Physically active sport science students; unclear
	RR group: n = 11	8/3	24 ± 2	175 ± 9	75 ± 11	Physically active sport science students; unclear
Rooney et al. (1994) [88]	TS group: n = 13	Unclear	18 – 35 ^b	Unclear	Unclear	Healthy young men; unclear
	CS group: n = 14	Unclear	18 – 35 ^b	Unclear	Unclear	Healthy young men; unclear
Goto et al. (2005) [34]	TS group: n = 9	9/0	23 ± 1	172 ± 2	65 ± 3	Healthy young men; unclear
	CS group: n = 9	9/0	22 ± 2	170 ± 2	66 ± 2	Healthy young men; unclear
Hansen et al. (2011) [42]	TS group: n = 9	9/0	25 ± 6	189 ± 10	107 ± 7	Elite rugby union athletes; unclear
	RR group: n = 9	9/0	28 ± 5	189 ± 10	100 ± 11	Elite rugby union athletes; unclear
Iglesias-Soler et al. (2016) [61]	TS group: n = 13	7/6	23 ± 5	173 ± 7	67 ± 9	Physically active sport science students; > 6 months
	RR group: n = 13	7/6	23 ± 5	173 ± 7	67 ± 9	Physically active sport science students; > 6 months
Iglesias-Soler et al. (2017) [86]	TS group: n = 13	7/6	23 ± 5	173 ± 7	67 ± 9	Physically active sport science students; > 6 months
	RR group: n = 13	7/6	23 ± 5	173 ± 7	67 ± 9	Physically active sport science students; > 6 months
Lawton et al. (2004) [33]	TS group: n = 13	13/0	Unclear	Unclear	Unclear	Junior elite athletes; > 6 months
	RR group: n = 13	13/0	Unclear	Unclear	Unclear	Junior elite athletes; > 6 months
Morales-Artacho et al. (2018) [90]	TS group: n = 9	9/0	24 ± 6 ^a	178 ± 6	77 ± 10	Physically active sport science students; no experience
	CS group: n = 9	9/0	24 ± 6 ^a	178 ± 6	77 ± 10	Physically active sport science students; no experience
Nicholson et al. (2016) [56]	TS group: n = 11	11/0	Unclear	Unclear	Unclear	Resistance-trained men; > 12 months
	CS group: n = 12	12/0	Unclear	Unclear	Unclear	Resistance-trained men; > 12 months
Oliver et al. (2013) [55]	TS group: n = 11	11/0	25 ± 5	180 ± 6	82 ± 12	Resistance-trained men; > 2 years
	RR group: n = 11	11/0	25 ± 4	180 ± 4	83 ± 10	Resistance-trained men; > 2 years
Rial-Vázquez et al. (2020) [89]	TS group: n = 13	10/3	23 ± 4 ^a	177 ± 8	73 ± 11	Physically active sport science students; > 3 months
	RR group: n = 11	7/4	23 ± 4 ^a	177 ± 8	73 ± 11	Physically active sport science students; > 3 months
Cuevas-Aburto et al. (2020) [57]	TS group: n = 12	12/0	21 ± 3	175 ± 5	73 ± 9	Physically active sport science students; unclear
	RR group: n = 13	12/0	20 ± 3	172 ± 5	75 ± 11	Physically active sport science students; unclear
Drinkwater et al. (2005) [32]	TS group: n = 11	15/0	18 ± 0 ^a	191 ± 9 ^a	86 ± 10 ^a	Junior elite athletes; > 6 months
	RR group: n = 15	11/0	18 ± 0 ^a	191 ± 9 ^a	86 ± 10 ^a	Junior elite athletes; > 6 months

CS cluster set structure, F female, M male, N, number of participants, RR rest redistribution set structure, TS traditional set structure

Values are expressed as means ± standard deviations

^aCombined for both groups

^bOnly range was reported

Table 2 Summary of the studies included in the review

Study	Training protocol (duration; sessions/w; exercise; intensity)	Structure Description Sets x repetitions; inter-set rest; intra-set rest	Control group trained to failure	Adherence	Outcomes; (methods of assessment)
Asadi et al. (2016) [58]	6 weeks; 2 sessions/w; 100 max intensity depth jumps from 45 cm height box per session	TS: 2×10; 120 s; none RR: 2×10; 90 s; 30 s	No	Unclear	Countermovement jump height (Vertec); Standing long jump height (measurement tape); t-test; 20 m sprint time; 40 m sprint time (all photocell gates)
Carneiro et al. (2019) [59]	8 weeks; 2 sessions/w (no consecutive days); unilateral leg extension (one leg performed TS, while the other leg performed CS); 90% 1RM	TS: 3×4; 90 s; none CS: 3×4; 90 s; 30 s	Yes	Unclear	Maximal strength (1RM unilateral leg extension); Thigh CSA (anthropometric equation (thigh circumference – $\pi \times \text{skinfold})/2/4\pi$); Force–velocity profiles, peak force, velocity and power (linear encoder)
Davies et al. (2019) [60]	8 weeks; 2 upper-body sessions and 1 lower-body session/w for first 4 weeks, 2 upper-body and 2 lower-body sessions/w for last 4 weeks. Set structures were manipulated only for bench press exercise with relative loads ranging from 70–75% of 1RM	TS: 4×5; 300 s; none CS: 4×5; 180 s; 30 s	No	Unclear	Maximal strength (1RM bench press); Mean and peak bench press velocity and power (linear position transducer)
Dias et al. (2020) [79]	12 weeks; 2 sessions/w; Bench press, leg press, seated row, deadlift, medicine ball throw, plank; 75% of 10RM for strength exercises, whereas medicine ball throw and the plank were performed for maximum distance and maximal hold, respectively	TS: 3×8; 120 s; none RR: 12×2; 30 s; none	No	At least 80%	Maximal strength (10RM Leg press and bench press); Medicine ball throw distance (measurement tape); Countermovement, squat jump and standing long jump height (contact mat); Muscular endurance (number of repetitions s in 30 s with 2 kg load)
Farinas et al. (2019) [80]	5 weeks; 2 sessions/w; biceps curl; 10RM load	TS: 5×6; 120 s and 15 s; none RR: 30×1; 18.5 s; none	No	Unclear	Maximal strength (1RM biceps curl); Muscular endurance (number of repetitions with 10RM load); Muscle thickness (Ultrasound)
Rooney et al. (1994) [88]	6 weeks; 3 sessions/w biceps curl; 6RM load	TS: 1×6–10; none; none CS: 1×6–10; none; 30 s	Yes	Unclear	Dynamic and isometric maximal strength (1RM biceps curl and strain-gauge force transducer, respectively)
Goto et al. (2005) [34]	12 weeks; 2 sessions/w, lat pulldown, shoulder press, bilateral knee extension; 75% 1RM	TS: 3–5×10; 60 s; none CS: 3–5×10; 60 s; 30 s	Yes	Unclear	Dynamic, isometric and isokinetic maximal strength (1RM shoulder press and knee extension, with only knee extension being measured isokinetically); Muscle endurance (max number of reps with 70% 1RM shoulder press and knee extension); Muscle CSA (magnetic resonance imaging)

Table 2 (continued)

Study	Training protocol (duration; sessions/w; exercise; intensity)	Structure Description Sets x repetitions; inter-set rest; intra-set rest	Control group trained to failure	Adherence	Outcomes; (methods of assessment)
Hansen et al. (2011) [42]	8 weeks; 2 sessions/w; front squat, back squat, clean pull, box squat, power clean, jump squat, with set structures being manipulated only for the 2-compound strength and power movements; 80 to 95% 1RM for the squat and clean pull movements during the entire program. In weeks 7 and 8, jump squats at 0–20% 1RM were also performed together with power cleans which replaced the clean pull exercise	TS: 3–5 × 4–8; 180 s; none RR: 3–12 × 1–6; 120–180 s; 10–30 sec ^a	Yes	Unclear	Maximal strength (1RM back squat prediction trough the repetitions to failure test with 2–6RM load); Peak velocity, peak power and peak force production during the back squat (portable force plate)
Iglesias-Soler et al. (2016) [61]	5 weeks; 2 sessions/w; unilateral leg extensions (TS with one leg and CS with the other leg); 10RM	TS: 4 × 8; 180 s; none RR: 31 × 1; 17.4 s; none	No	Unclear	Dynamic max strength (1RM leg extension); isometric max strength (leg extension with force cell); Muscular endurance (time to task failure with 50% of MVC); Thigh girth (corrected girth = thigh girth – ($\pi \times$ thigh skinfold)); Rate of force development (during MVC with force cell); Mean propulsive power (linear velocity transducer)
Iglesias Soler et al. (2017) [86]	5 weeks; 2 sessions/w; unilateral leg extensions (TS with one leg and CS with the other leg); 10RM	TS: 4 × 8; 180 s; none RR: 31 × 1; 17.4 s; none	No	Unclear	Force–velocity profiles (linear velocity transducer)
Lawton et al. (2004) [33]	6 weeks; 3 sessions/w (alternate days); 3 1RM intensity was increased by 5% over the last 3 weeks of the training intervention	TS: 4 × 6; 240 s and 20 s; none RR: 8 × 3; 60 s and 53 s; none	Yes	Unclear	Maximal strength (6RM bench press); Mean power during bench press throws (linear encoder)
Morales-Artacho et al. (2018) [90]	11 weeks; 2 sessions/w (nonconsecutive days); loaded countermovement squat jumps on a Smith machine at 20% of 1RM and abdominal plank	TS: 6 × 6; 300 s; none CS: 6 × 6; 240 s and 30 s; 30 s	No	100%	Force–velocity profiles in the counter movement jump, peak force, velocity and power (linear encoder)
Nicholson et al. (2016) [56]	6 weeks; 2 sessions/w; back squat; 85% 1RM	TS: 4 × 6; 300 s; none CS: 4 × 6; 300 s; 25 s	Yes	Unclear	Dynamic (1RM back squat), isometric (1RM back squat test on force plate) and isokinetic strength (dynamometer); Counter movement jump height (force plate)

Table 2 (continued)

Study	Training protocol (duration; sessions/w; exercise; intensity)	Structure Description Sets x repetitions; inter-set rest; intra-set rest	Control group trained to failure	Adherence	Outcomes; (methods of assessment)
Oliver et al. (2013) [55]	12 weeks; 2 sessions/w; bench press, back squat; 60 to 75% 1RM	TS: 4 × 10; 120 s; none RR: 8 × 5; 60 s; none	No	95%	Dynamic muscular strength (1RM back squat and bench press); Mean power during back squat and bench press (linear position transducer) as well as during counter movement jumping (determined based on jump height measured using Vertec); Lean body mass (DEXA)
Rial-Vázquez et al. (2020) [89]	5 weeks; 2 sessions/w; bench press, back squat, lat pulldown, leg curl; 10RM	TS: 4 × 8; 300 s; none RR: 16 × 2; 60 s; none	No	Unclear	Dynamic muscular strength (1RM back squat); Force–velocity profile in the back squat (linear velocity transducer); Maximal propulsive power in the back squat and bench press (linear velocity transducer)
Cuevas-Aburto et al. (2020) [57]	6 weeks; 2 sessions/w; bench press; 75% of 1RM	TS: 6 × 5; 180 s; none RR: 30 × 1; 31 s; none	No	100%	Dynamic muscular strength (1RM bench press); Peak velocity during bench press throws (linear velocity transducer); handball throwing velocity (radar gun)
Drinkwater et al. (2005) [32]	6 weeks; 3 sessions/w (alternate days); 3 1RM intensity was increased by 5% over the last 3 weeks of the training intervention	TS: 4 × 6; 240 s and 20 s; none RR: 8 × 3; 60 s and 53 s; none	Yes	Unclear	Dynamic muscular strength (6RM bench press); Mean power during bench press throws (linear encoder)

CS, cluster set structure, CSA cross-sectional area, DEXA dual-energy x-ray absorptiometry, MVC maximum voluntary contraction, RM Repetition maximum, RR rest redistribution set structure, TS traditional set structure,

Values are expressed as means ± standard deviations

^aOnly range was reported

high and downgraded one level to moderate, low, or very low for each of the following limitations: imprecision, inconsistency, and risk of bias. For imprecision, the level of evidence was downgraded one level if the conclusion about the effect magnitude would be altered based on the lower or upper boundary of the confidence interval. For example, if the mean effect was small, but the 95% confidence interval crossed the threshold for a trivial effect (i.e., $g < 0.2$), the precision was insufficient to support a strong recommendation as the confidence interval does not exclude the possibility for the effect to be trivial. Similarly, if the confidence interval crossed the threshold for a large effect, while the mean effect was moderate, the conclusion was considered imprecise and as such, the level of evidence was also downgraded one level. Finally, if high statistical heterogeneity was observed, and if more than 50% of studies had > 1 risk of bias item assessed to be high risk, the level of evidence was also downgraded one level for inconsistency and risk of bias, respectively.

2.6 Statistical Analysis

When two or more studies reported on the same outcome, a separate random-effects meta-analysis for each outcome was performed using the Meta and Metafor statistical packages [66, 67] in R language and environment for statistical computing (version 3.6.3, The R foundation for Statistical Computing, Vienna, Austria) [68]. Model parameters were calculated by the restricted maximum likelihood method, and the observations were weighted by the inverse of the sampling variance [69]. Sub-group analyses were used to explore the effects of CS and RR set structures on every outcome. Meta-regressions were performed when at least six effects were available for a given outcome [70]. These included the effects of TS performed to muscular failure as a categorical covariate, and study duration, loads, and the number of repetitions performed in the TS group as continuous covariates.

The heterogeneity of the studies was assessed using the I^2 statistic. The I^2 statistic represents the percentage of total variation in estimated effects across studies due to heterogeneity rather than chance and was interpreted as small ($I^2 < 25\%$), moderate ($I^2 = 25\text{--}49\%$) and high ($I^2 > 50\%$) [71]. Publication bias was not assessed because there was only a small number of studies included in most analyses and all studies were of a similar sample size, as recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [72]. In addition, there was no

reason to expect that studies finding no significant difference between different set structures would be less likely to be published than studies reporting a statistically significant difference.

2.6.1 Determination of the Effect Size

The standardised mean change for each group was calculated as the difference between post-test and pre-test scores, divided by the pre-test standard deviation with an adjustment for small sample bias [73–75]. Thereafter, the standardised mean difference (SMD) with 95% confidence intervals for the difference were calculated between the two standardised mean changes values for each study to indicate how much larger (or smaller) the change in the experimental group was (in standard deviation units) when compared to the change in the control group [73, 74]:

$$\text{SMD} = C_{\text{Exp}} \left(\frac{M_{\text{post, Exp}} - M_{\text{pre, Exp}}}{\text{SD}_{\text{pre, Exp}}} \right) - C_{\text{Con}} \left(\frac{M_{\text{post, Con}} - M_{\text{pre, Con}}}{\text{SD}_{\text{pre, Con}}} \right)$$

where C (i.e. the correction for small sample bias) is given by:

$$C_j = 1 - \frac{3}{4(n_j - 1) - 1}$$

The SMD magnitude was interpreted as: small (0.20 – 0.49), moderate (0.50 – 0.79) or large (> 0.80) [76].

2.6.2 Determination of the Variance

No studies reported the pre- to post-intervention correlations required to determine the variance. Therefore, when the authors did not provide correlations upon our request, standard deviations of the pre- to post-intervention change were used to calculate pre-to-post correlations using the following formula [77]:

$$r_j = \frac{\text{SD}_{j,\text{baseline}}^2 + \text{SD}_{j,\text{final}}^2 - \text{SD}_{j,\text{change}}^2}{2 * \text{SD}_{j,\text{baseline}} * \text{SD}_{j,\text{final}}}$$

The corresponding authors were contacted when the standard deviations of the pre- to post-intervention change were not reported. If it was not possible to obtain standard deviation of the pre- to post-intervention change, the median correlation from all other studies for a given outcome was imputed. This ensured that the maximum number of studies were included. Since the experimental and control groups are independent, the corresponding sampling variances were

calculated by summing the sampling variances of the two groups [73].

2.6.3 Combining Effects and Further Decisions

When multiple study effects were reported that were not of a direct interest to this review (e.g., separate effects for two or more TS, CS, or RR set structures), a combined effect was computed across the outcome for each group separately (i.e., control and experimental), as outlined by Boronstein et al. [78] for dependent continuous outcomes. This avoided double counting individuals from those studies in the meta-analysis.

While extracting the data from the studies, we noticed that several studies [34, 61, 79, 80] reported the effects of alternative set structures on muscular endurance as assessed by the work completed during repetitions to muscular failure. Therefore, although not originally planned we also meta-analysed the effects of alternative set structures on muscular endurance. Furthermore, while we planned to analyse the effects of alternative set structures on the overall expression of muscular power, which included changes in jump height, sprint and change of direction times as well as velocity and power outputs at submaximal loads, this was not done due to a lack of studies reporting on all these outcomes. Rather, we separately meta-analysed vertical jump performance and velocity and power outputs at submaximal loads due to a large number of studies that reported these outcomes. For vertical jump performance, only unloaded jump assessments were considered while only loaded exercises (both ballistic and strength-oriented) were considered for velocity and power outputs at submaximal loads. If studies reported multiple measures of muscular strength (i.e., using 1RM, MVC, or isokinetic testing), tests with a RM component were used for analysis as these are generally considered more ecologically valid. Similarly, when studies reported changes in both velocity and power outputs at submaximal loads after the intervention, only velocity was used for the analysis. This was done because velocity: (a) is more stable and reliable than power regarding changes in performance [81, 82]; and (b) has a strong relationship with fatigue and mechanical impulse, which are highly related to various athletic tasks' performance [19, 22, 83]. In addition, we used mean velocity and power variables for the non-ballistic exercises whereas peak variables were used for the ballistic exercises, where possible. This was done since peak velocity and power variables are pertinent measures for explosive-type resistance training while mean variables are believed to better represent velocity (and are almost exclusively used for this purpose) velocity through the entire phase of non-ballistic movements

[84, 85]. In three studies [59, 61, 86], the unilateral knee extension exercise was employed, and participants' legs were randomly assigned to TS and RR groups. We considered these groups to be independent since previous studies have shown no correlation between legs (left and right) for strength gains after different RT protocols, suggesting that when both limbs are trained, the cross-education effect is minimal [87]. In four studies [32, 42, 55, 88], resistance training volume or intensity were periodised across microcycles (i.e., weeks) and mesocycles (i.e., 4-week blocks). In this case, the average number of repetitions performed and the average intensity of load were calculated to simplify the analyses. When studies did not specify whether the repetitions in the control group (i.e., TS) were performed to muscular failure, the study protocol was classified as "to failure" when some of the participants either failed to complete the protocol or when the combination of the number of repetitions and the load clearly suggested that to be the case (e.g. doing more than 5 repetitions with 85% 1RM). Finally, to support readability, all effects were expressed relative to TS, such that positive values represent outcomes in favour of CS or RR.

3 Results

3.1 Search Results

The primary search yielded 1275 results, of which 8 studies met the inclusion criteria. Forward citation tracking of the included studies as well as monitoring newly published, relevant literature yielded an additional 9 studies that met the inclusion criteria, resulting in 17 studies included in this review. The stages of the search and study selection process is presented in Fig. 1.

3.2 Study Characteristics

The total number of participants pooled across studies was 418 (249 males and 129 females with 40 not specified). However, after thorough inspection of the studies, it was observed that the data from the same participants were used in multiple studies [32, 33, 61, 86]. Therefore, the total number of participants was reduced to 366 (209 males, 117 females, and 40 unspecified). Of the seventeen included studies, two studies included only females [59, 79], five studies [60, 61, 80, 86, 89] included males and females, two studies did not specify sex [58, 88], and the remaining studies included only males. Participants in eight studies had at least 6 months of resistance training experience, in three studies participants had no previous resistance training experience, while prior resistance training experience was not reported in six studies. Mean study duration was

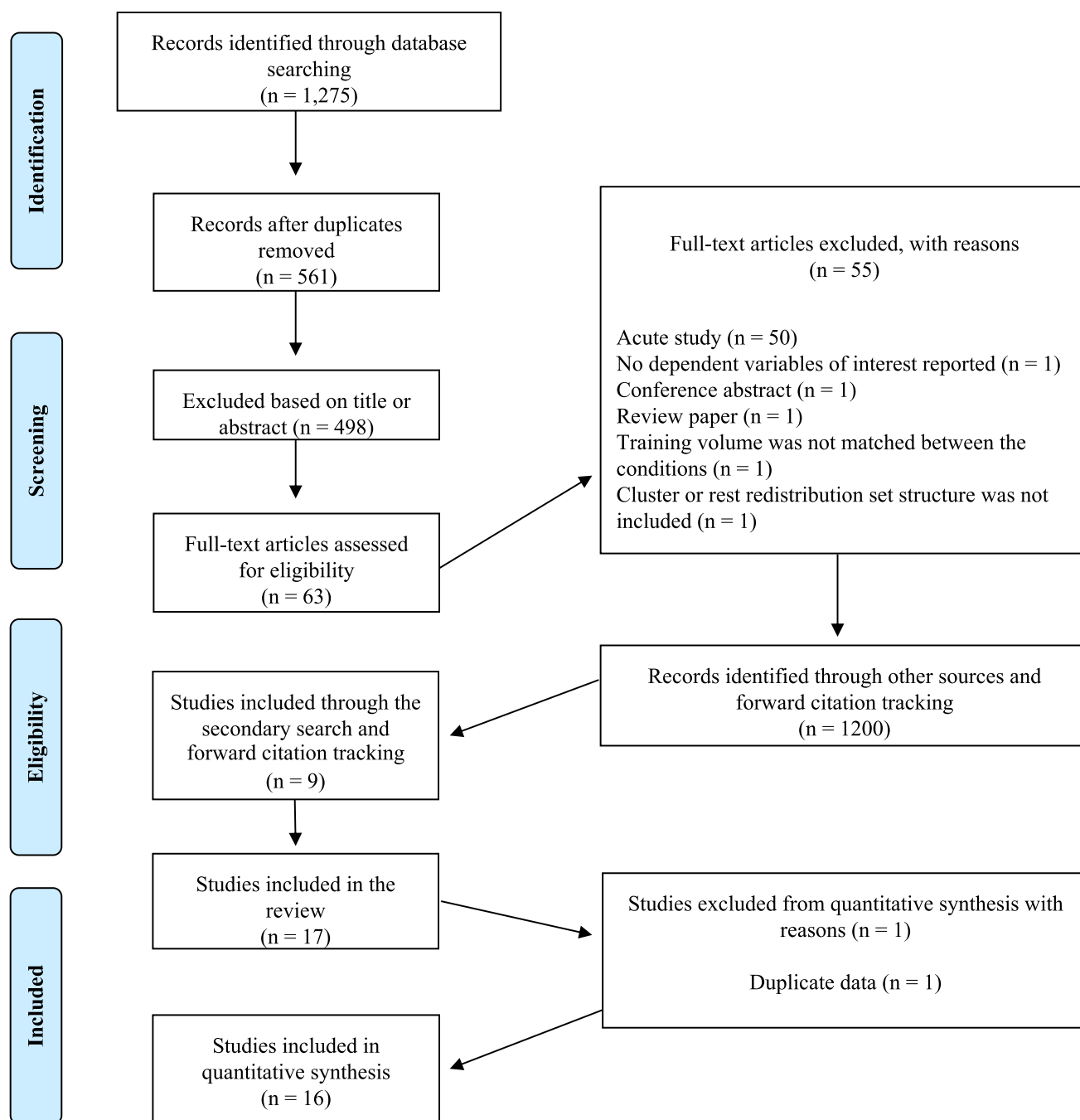


Fig. 1 Literature search flow chart. *n* number of studies

approximately 7 weeks (range 4–12 weeks). Only four studies specified adherence with the intervention [55, 57, 79, 90] which was generally very high. Ten studies compared RR to TS, and seven studies compared CS to TS. Exercises during training interventions were performed using exclusively weight machines in seven studies, Smith machines or

free-weights in three studies, while the combination of these training modalities was used in four studies. A more comprehensive description of the participants and the included studies can be found in Tables 1 and 2.

3.3 Risk of Bias Assessment

According to the risk of bias assessment, two studies [34, 56] were at high risk of an order effect bias since they did

not report any randomisation. Only one study [59] was classified as low risk, providing sufficient information regarding the method of randomisation. The remaining studies were classified as an unclear risk since they did not provide sufficient information regarding the method of randomisation. No studies provided information regarding outcome blinding procedures. One study [79] attempted to blind the investigators to group allocation and therefore was considered to be at a low risk of allocation concealment bias, whereas all other studies were rated as an unclear risk of bias due to lack of information regarding the allocation concealment. Two studies [88, 91] were rated at a high risk of attrition bias, excluding randomised participants from the analysis

without specifying the minimum amount of training sessions required for the participants to be included in the analysis or reporting study dropouts without sufficient reason. One study [56] did not provide sufficient information on the number of participants assessed and included in the analysis after reporting that some of them did not complete the entire intervention and hence, was classified as an unclear risk of attrition bias. No studies, except for one [79], pre-registered their protocols on a publicly available registry platform, so it was unclear whether selective reporting bias was present. One study included in this review pre-registered their protocol [79] and reported the majority (but not all) of the outcomes; however, the outcomes of interest to the present

Fig. 2 Risk of bias assessment for all included studies

Asadi et al. (2016) [58]	?	?	?	+	?	+	+	×
Carneiro et al. (2019) [59]	+	?	?	+	?	+	×	+
Cuevas–Aburto et al. (2020) [57]	?	?	?	×	?	+	+	+
Davies et al. (2019) [60]	?	?	?	+	?	×	+	+
Dias et al. (2020) [79]	?	?	+	+	+	×	+	+
Drinkwater et al. (2005) [32]	?	?	?	+	?	×	+	+
Farinas et al. (2019) [80]	?	?	?	+	?	×	+	+
Goto et al. (2005) [34]	×	?	?	+	?	+	+	+
Hansen et al. (2011) [42]	?	?	?	+	?	×	+	+
Iglesias–Soler et al. (2016) [61]	?	?	?	+	?	?	+	+
Iglesias–Soler et al. (2017) [86]	?	?	?	+	?	?	+	+
Lawton et al. (2004) [33]	?	?	?	+	?	×	+	+
Morales–Artacho et al. (2017) [90]	?	?	?	+	?	+	+	+
Nicholson et al. (2016) [56]	×	?	?	?	?	×	+	+
Oliver et al. (2013) [55]	?	?	?	+	?	×	+	+
Rial–Vázquez et al. (2020) [89]	?	?	?	+	?	+	+	+
Rooney et al. (1994) [88]	?	?	?	×	?	+	+	+
	Random sequence bias	Allocation concealment bias	Blinding of outcome assessment	Incomplete data	Selective reporting	Set structure bias	Outcome assessment bias	Familiarisation bias

Judgement
 × High
 + Low
 ? Unclear

review were reported. Therefore, the study was classified as low risk of selective reporting bias. Eight studies [32, 33, 42, 55, 56, 60, 79, 80] were considered high risk for set structure bias since the participants performed other RT sessions (in which set structures were not manipulated), or because they did not perform all exercises using a single-set structure, but rather a combination of set structures (e.g., CS for the back squat, TS for the accessory exercises). Information on set structure was unclear in two studies [61, 86]. Only one study [59] used a linear encoder that was not, to our knowledge, validated in peer-reviewed literature whereas all other studies used established methods, equipment, or instruments to evaluate their outcomes of interest. While four studies [32, 33, 42, 57] did not include a familiarisation session, two of them [58, 88] were classified as high risk of bias for not having a familiarisation session which might have affected the habituation to the study protocols. The rest of the studies performed a test–retest reliability of their assessment procedures, performed an acute study (i.e., one or two sessions) examining the short term effects of their protocol (with all participants) or specifically stated that all participants were accustomed to the study protocols (i.e., had performed them in the past). A more detailed risk of bias assessment is illustrated in Fig. 2.

3.4 Muscular Strength

Sub-group analysis revealed that neither CS (SMD = -0.07 [95% CI: $-0.21, 0.07$]; $p = 0.300$; $I^2 = 0\%$; $k = 5$) nor RR (SMD = -0.04 [95% CI: $-0.20, 0.12$]; $p = 0.641$; $I^2 = 51\%$; $k = 8$) set structures were more effective than TS structures in promoting muscular strength adaptations. Likewise, when the effects from alternative set structures were pooled, there were no significant differences (SMD = -0.06 [95% CI: $-0.16, 0.05$]; $p = 0.291$; $I^2 = 32\%$; $k = 13$) in muscular strength gains compared to TS structures (Fig. 3). Sub-group results for CS and RR provided moderate whereas pooled results provided high GRADE quality of evidence (Table 3).

Performing repetitions to muscular failure ($b = -0.22$ [95% CI: $-0.46, 0.16$]; $p = 0.067$), the number of repetitions performed during TS ($b = 0.05$ [95% CI: $-0.02, 0.11$]; $p = 0.146$), the load used ($b = -0.01$ [95% CI: $-0.03, 0.01$]; $p = 0.249$) and study duration ($b = 0.02$ [95% CI: $-0.04, 0.08$]; $p = 0.526$) were not significant moderators of the SMD for muscular strength adaptations (Fig. 4). Likewise, when analysing RR set structures separately, performing repetitions to muscular failure ($b = -0.45$ [95% CI: $-0.97, 0.07$]; $p = 0.094$) and the study duration ($b = 0.03$ [95% CI: $-0.04, 0.10$]; $p = 0.397$) were not significant moderators of the SMD for muscular strength gains. However, the number of repetitions performed during TS ($b = 0.14$ [95% CI: $0.05, 0.24$]; $p = 0.035$) and the load used ($b = -0.05$ [95% CI:

Fig. 3 Random-effects meta-analysis of the differences in muscular strength gains after resistance training using alternative set structures (i.e., CS and RR) compared to TS. Subgroup analysis based on the effects of CS and RR separately. CI confidence interval, CS cluster set structure, RR rest redistribution set structure, SMD standardised mean difference, TS traditional set structure

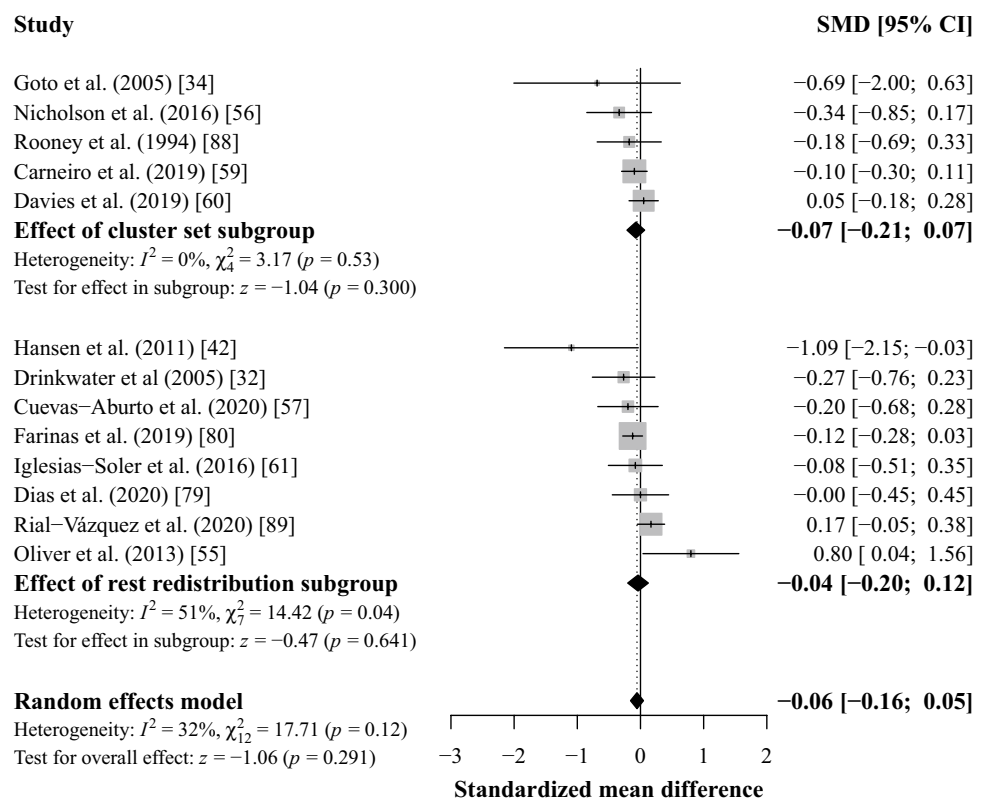


Table 3 Summary of meta-analysis findings and quality of evidence synthesis

Outcome	Summary of findings				Quality of evidence synthesis (GRADE)			
	<i>k</i>	<i>n</i>	Effect (95% CI)	Direction of the effect compared to traditional sets	Imprecision	Inconsistency	Risk of bias	Overall quality
Muscular strength								
Cluster set	5	120	− 0.07 (− 0.21, 0.07)	↔	− 1	None	None	Moderate
Rest redistribution	8	230	− 0.04 (− 0.20, 0.12)	↔	− 1	None	None	Moderate
All	13	350	− 0.06 (− 0.16, 0.05)	↔	None	None	None	High
Hypertrophy								
Cluster set	2	49	− 0.93 (− 3.24, 1.38)	↔	− 1	− 1	None	Low
Rest redistribution	3	71	− 0.06 (− 0.26, 0.14)	↔	− 1	None	None	Moderate
All	5	120	− 0.03 (− 0.20, 0.13)	↔	− 1	− 1	None	Low
Muscular endurance								
Rest redistribution	3	115	− 0.37 (− 0.72, − 0.02)	↓	− 1	None	None	Moderate
All	4	133	− 0.38 (− 0.69, − 0.07)	↓	− 1	None	None	Moderate
Vertical jump performance								
Rest redistribution	4	119	0.16 (− 0.09, 0.40)	↔	− 1	None	None	Moderate
All	5	142	0.13 (− 0.07, 0.33)	↔	− 1	None	None	Moderate
Velocity and power at sub-max loads								
Cluster set	4	93	0.19 (− 0.01, 0.40)	↔	− 1	None	None	Moderate
Rest redistribution	7	207	0.16 (− 0.04, 0.37)	↔	− 1	None	None	Moderate
All	11	300	0.18 (0.04, 0.31)	↑	− 1	None	None	Moderate
Slope of the force–velocity profile								
Cluster set	2	49	0.30 (0.13, 0.47)	↑	− 1	None	None	Moderate
Rest redistribution	2	50	0.17 (− 0.30, 0.65)	↔	− 1	None	None	Moderate
All	4	99	0.29 (0.13, 0.44)	↑	− 1	None	None	Moderate

CI confidence intervals, *GRADE* Grading of Recommendations Assessment, Development and Evaluation, *k* number of studies, *n* number of participants

Only outcomes with *k* > 1 are included in this table

↑ More effective when compared to traditional set structures

↔ Equally effective when compared to traditional set structures

↓ Less effective when compared to traditional set structures

− 0.10, − 0.01]; *p* = 0.028) were significant moderators of the SMD in muscular strength gains, with higher number of repetitions during TS and lower loads used leading to greater effectiveness of RR on muscular strength gains.

3.5 Hypertrophy

Sub-group analysis indicated that neither CS (SMD = − 0.93 [95% CI: − 3.24, 1.38]; *I*² = 87%; *p* = 0.429; *k* = 2) nor RR (SMD = − 0.06 [95% CI: − 0.26, 0.14]; *p* = 0.543; *I*² = 0%; *k* = 3) set structures were less effective in promoting muscle hypertrophy. Pooling the effects from alternative set structures resulted in no significant differences (SMD = − 0.03

[95% CI: − 0.20, 0.13]; *p* = 0.708; *I*² = 52%; *k* = 5) in muscle hypertrophy when compared to TS structures. Sub-group results for CS provided low, whereas sub-group results for RR and pooled results provided moderate and low GRADE quality of evidence, respectively (Table 3).

3.6 Muscular Endurance

Sub-group analysis indicated that RR (SMD = − 0.37 [95% CI: − 0.72, − 0.02]; *p* = 0.038; *I*² = 0%; *k* = 3) but not CS set structures (SMD = − 0.42 [95% CI: − 1.07, 0.23]; *p* = 0.203; *I*² = 0%; *k* = 1) were less effective at improving muscular endurance than TS structures. When the effects

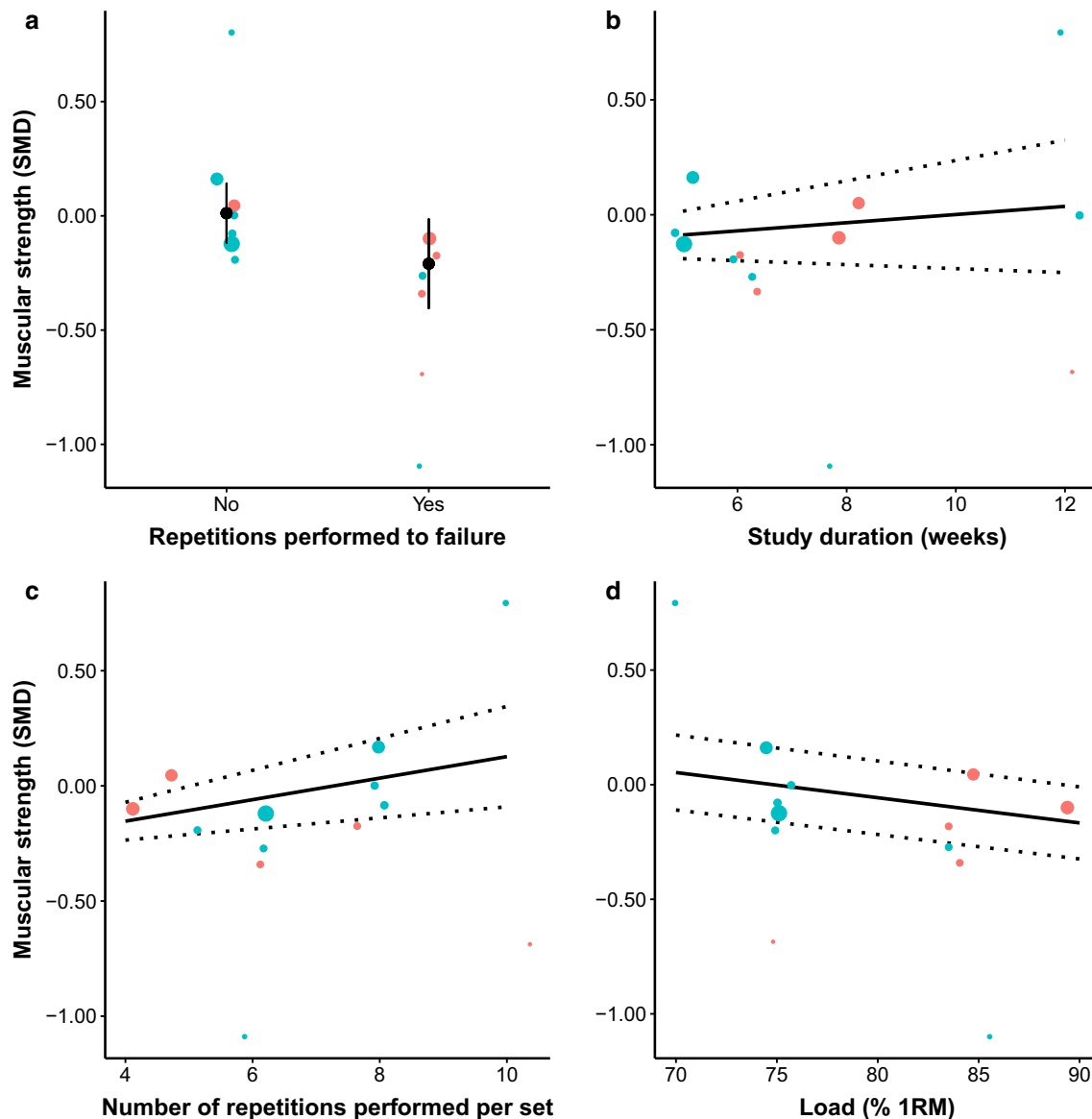


Fig. 4 Random-effects meta-regression of the differences in muscular strength gains after resistance training using alternative set structures (i.e., CS and RR) compared to TS while controlling for the effects of muscular failure during TS (**a**), study duration in weeks (**b**), number of repetitions performed during TS (**c**) and the load used (**d**). Larger data points received greater weighting than smaller data points. Red

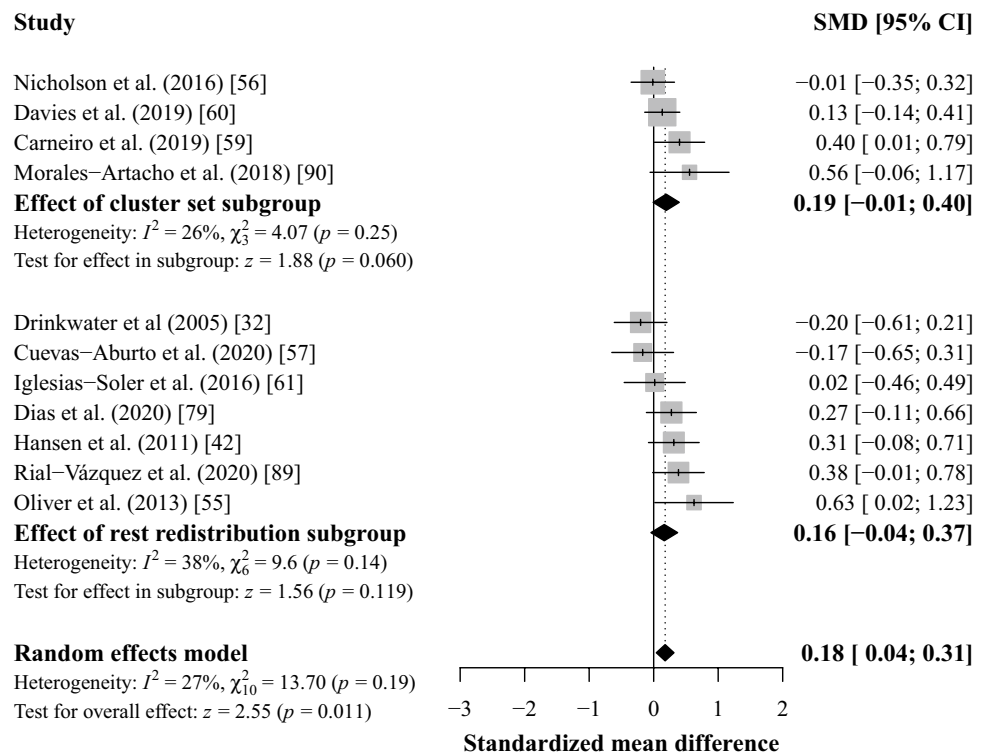
and blue circles represent studies that compared TS with CS and RR, respectively. Solid (non-vertical) lines represent the estimated relationship, solid vertical lines and dotted lines represent the upper and lower 95% confidence intervals. *CS* cluster set structure, *RM* repetition maximum, *RR* rest redistribution set structure, *TS* traditional set structure

from alternative set structures were pooled, they were less effective (SMD = -0.38 [95% CI: -0.69, -0.07]; $p = 0.015$; $I^2 = 0\%$; $k = 4$) at improving muscular endurance compared to TS structures. Pooled results and sub-group results for RR provided moderate GRADE quality of evidence (Table 3).

3.7 Vertical Jump Performance

Sub-group analysis revealed that neither CS (SMD = 0.11 [95% CI: -0.37, 0.59]; $p = 0.659$; $I^2 = 0\%$; $k = 1$) nor RR (SMD = 0.16 [95% CI: -0.09, 0.40]; $p = 0.208$; $I^2 = 0\%$; $k = 4$) were more effective at improving vertical jump performance compared to TS structures. Likewise, when the effects from alternative set structures were pooled, both alternative and TS structures were equally effective (SMD = 0.13 [95% CI:

Fig. 5 Random-effects meta-analysis of the differences in velocity and power at submaximal loads after resistance training using alternative set structures (i.e., CS and RR) compared to TS. Subgroup analysis based on the effects of CS and RR separately. *CI* confidence interval, *CS* cluster set structure, *RR* rest redistribution set structure, *SMD* standardised mean difference, *TS* traditional set structure



-0.07, 0.33]; $p=0.190$; $I^2=0\%$; $k=5$) at improving vertical jump performance. Pooled results and sub-group results for RR provided moderate GRADE quality of evidence (Table 3).

3.8 Velocity and Power Outputs at Submaximal Loads

Sub-group analysis revealed that neither CS (SMD=0.19 [95% CI: -0.01, 0.40]; $p=0.060$; $I^2=26\%$; $k=4$) nor RR (SMD=0.16 [95% CI: -0.04, 0.37]; $p=0.119$; $I^2=38\%$; $k=7$) were more effective at inducing greater velocity and power outputs at submaximal loads compared to TS structures. However, when the effects from alternative set structures were pooled, significantly greater velocity and power outputs at submaximal loads (SMD=0.18 [95% CI: 0.04, 0.31]; $p=0.011$; $I^2=27\%$; $k=11$) were observed after interventions utilising alternative set structures compared to TS structures (Fig. 5). Pooled results and sub-group results for CS and RR provided moderate GRADE quality of evidence (Table 3).

Performing repetitions to muscular failure ($b=-0.10$ [95% CI: -0.39, 0.20]; $p=0.510$), the number of repetitions performed during TS ($b=0.04$ [95% CI: -0.06, 0.14]; $p=0.392$) as well as the load used ($b=-0.01$ [95% CI: -0.02, 0.00]; $p=0.236$), was not significant moderators of the SMD for enhancing velocity and power outputs at submaximal loads (Fig. 6). However, study duration was a significant moderator of the SMD ($b=0.06$ [95% CI: 0.01,

0.11]; $p=0.037$) with longer duration studies showing greater effectiveness for alternative set structures in enhancing velocity and power outputs at submaximal loads over TS structures (Fig. 6). When analysing RR set structures separately, performing repetitions to muscular failure ($b=-0.15$ [95% CI: -0.63, 0.32]; $p=0.525$), the number of repetitions performed during TS ($b=0.12$ [95% CI: -0.02, 0.25]; $p=0.087$) as well as the load used ($b=-0.02$ [95% CI: -0.06, 0.03]; $p=0.525$) and the study duration ($b=0.05$ [95% CI: -0.02, 0.12]; $p=0.175$), was not significant moderators of the SMD in enhancing velocity and power outputs at submaximal loads.

3.9 Slope of Force-Velocity Profiles

Sub-group analysis revealed that CS (SMD=0.30 [95% CI: 0.13, 0.47]; $p<0.001$; $I^2=0\%$; $k=2$) but not RR set structures (SMD=0.17 [95% CI: -0.30, 0.65]; $p<0.480$; $I^2=22\%$; $k=2$) promoted greater shifts of the slope of force-velocity profiles towards a more velocity dominant profile compared to TS structures. Likewise, when the effects from alternative set structures were pooled, greater shifts towards velocity dominant profiles were seen (SMD=0.29 [95% CI: 0.13, 0.44]; $p<0.001$; $I^2=0\%$; $k=4$) compared to TS structures. Pooled results and sub-group results for CS and RR provided moderate GRADE quality of evidence (Table 3).

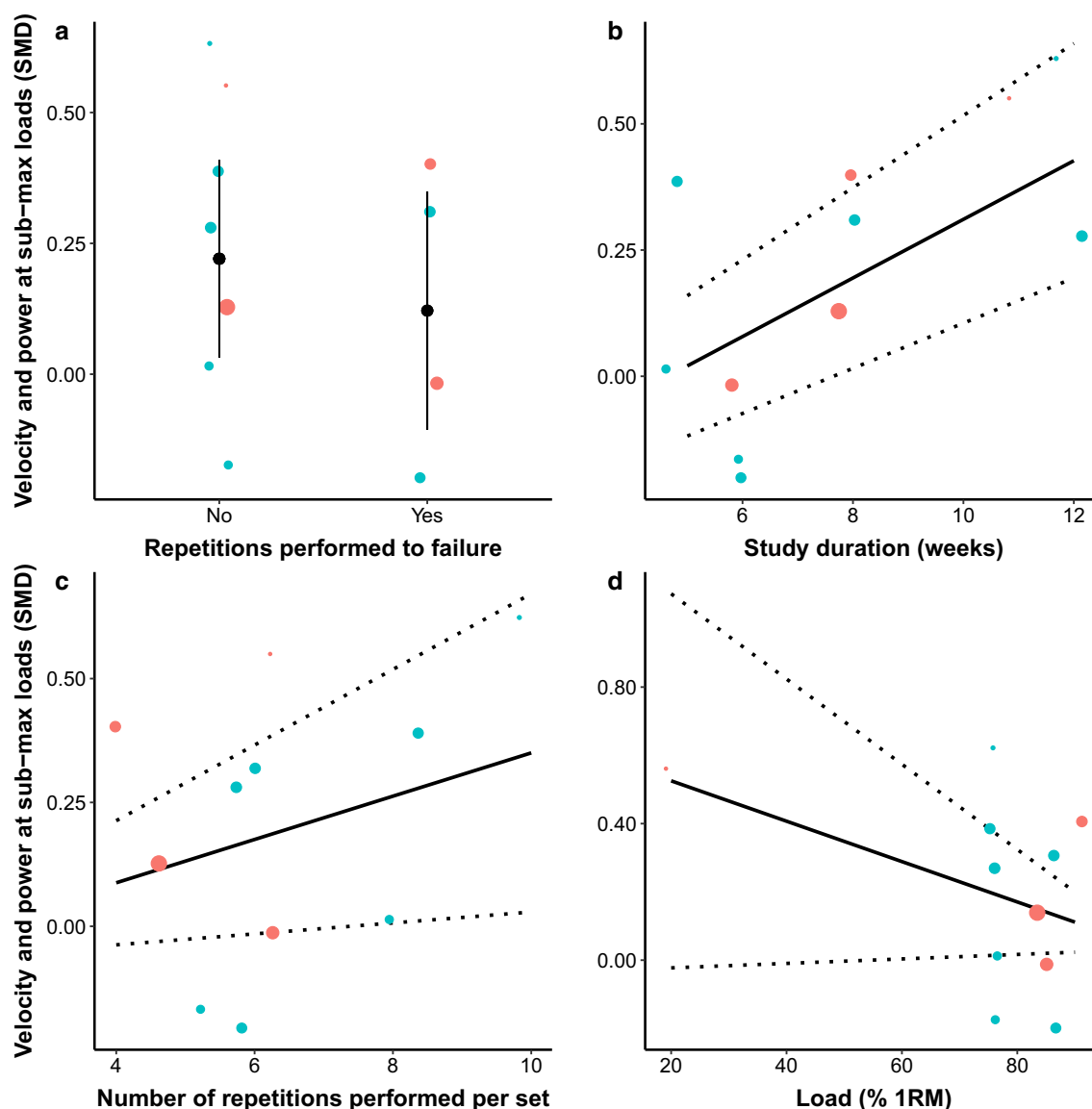


Fig. 6 Random-effects meta-regression of the differences in velocity and power at submaximal loads after resistance training using alternative set structures (i.e., CS and RR) compared to TS while controlling for the effects of muscular failure during TS (**a**), study duration in weeks (**b**), number of repetitions performed during TS (**c**) and the load used (**d**). Larger data points received greater weighting than

smaller data points. Red and blue circles represent studies that compared TS with CS and RR, respectively. Solid (non-vertical) lines represent the estimated relationship, solid vertical lines and dotted lines represent the upper and lower 95% confidence intervals. CS cluster set structure, RM repetition maximum, RR rest redistribution set structure, TS traditional set structure

4 Discussion

The present systematic review and meta-analysis sought to compare the effects of TS, CS, and RR in promoting various training adaptations. The findings of this review suggest that on a volume-equated basis: (1) set structure manipulation during RT has a negligible impact on muscular strength gains and hypertrophic outcomes; (2) TS is more effective at promoting muscular endurance gains; (3) alternative set structures are more effective at inducing greater velocity and power outputs at submaximal loads; (4) manipulation

of the set structure can modulate force–velocity profiles with alternative set structures leading to significant shifts towards velocity-oriented profiles; and (5) while the pooled SMD favoured alternative set structure for vertical jump performance gains, this effect was not statistically significant. Therefore, modifying TS to an alternative set structure (CS or RR) can influence the magnitude of specific muscular adaptations, which further suggests set structure manipulation to be an important consideration for RT program design.

4.1 Effects of Set Structure Manipulation on Muscular Strength, Hypertrophy, and Endurance

Moderate-to-high GRADE quality evidence indicates that set structure manipulation during RT has negligible effects on muscular strength gains and hypertrophic (low to moderate GRADE) outcomes when total training volume is equated. More specifically, TS was not more effective than CS or RR at increasing strength or promoting muscle hypertrophy. High levels of effort often experienced during TS are associated with increased metabolic by-product accumulation, motor unit recruitment, and endogenous hormone secretion, all of which have been proposed to contribute to muscle hypertrophy and enhanced muscle strength [23, 92]. Indeed, two studies [34, 42] that reported the greatest advantage of TS over CS and RR used this reasoning to explain their findings. In contrast to this, Oliver et al. [55] found RR to be equally effective in promoting muscle growth as TS, but superior for increasing muscle strength. The explanation for this finding may be found in the study by Burd et al. [92] who demonstrated that maximal electromyographic activity occurred half way through (i.e., repetitions 5–6) the performance of a set of leg extensions using a load corresponding to 70% 1RM. Since Oliver et al. [55] used similar intensity range and divided the number of repetitions performed during TS (i.e., 4×10) in half (i.e., 8×5) and equated total rest time, this may have resulted in greater neuromuscular activation over consecutive sets compared with TS, further contributing to the greater strength gains in the RR group. That said, there is an ongoing debate whether significant amounts of fatigue during RT are needed to optimise strength and hypertrophy adaptations.

More recent studies suggest that terminating sets after reaching only moderate levels of effort can lead to similar or greater strength gains when compared to highly fatiguing RT protocols [21, 38–40]. While training to muscular failure was not a significant moderator ($p > 0.067$) of the differences between TS and alternative set structures in the present review, there was still some indication of greater effectiveness of alternative set structures on strength gains when TS are performed to failure. In addition, based on the available evidence [93], stopping several repetitions short of failure when training with moderately heavy loads does not seem to compromise hypertrophy, at least when training volumes are equated. The findings of the present review, therefore, support these conclusions since TS (typically performed near or to muscular failure) and CS or RR (typically performed further from muscular failure), were equally effective at promoting muscular strength and hypertrophy adaptations. Although equally effective, it is plausible that the underlying mechanisms for these improvements are different for each set configuration. For instance, it is well established that CS

and RR allow for greater velocity and power outputs than TS whereas higher metabolic stress and perceived exertion can be seen after TS [54, 94]. Therefore, mechanical factors may have more relevance in the adaptations to RT utilising CS and RR whereas metabolic stimuli may be a predominant factor after traditional RT [95].

Moderate GRADE quality of evidence suggests that alternative set structures are less effective at improving muscular endurance than TS. These findings are not surprising since local muscular endurance was examined using the repetitions to failure task with a submaximal load in all studies included in the present review. It is important to note that muscular abilities, such as endurance, have a significant skill component, meaning that individuals are likely to learn the specific pattern of muscle recruitment associated with performance of a given task [96]. Indeed, in a study by Mattocks et al. [97], one group trained with a repetition range that mimicked the strength test, while the other group trained with a repetition range similar to the endurance test, and subsequently performed better in the muscular endurance test (although not in all muscle groups). Therefore, greater muscular endurance adaptations are probably influenced by the greater task-dependent similarities between the testing procedures and TS, rather than CS or RR. It is plausible that muscular endurance adaptations assessed through an intermittent task (e.g., repeated power ability) would be greater after RT using alternative set structures, but this remains to be investigated. Most of the studies examining the effects of set structure manipulation on local muscular endurance have analysed lower-body exercises and found superior effects of TS over CS and RR. Only Goto et al. [34] analysed both a lower-body and upper-body exercise, and reported TS to be more effective than CS only for the knee extension but not shoulder press exercise, suggesting that the differences in local muscular endurance adaptations between TS and alternative set structures might be exercise specific. While previous studies suggested that upper-limb muscles may have a greater trainability than lower-limb muscles that are more involved in daily physical activities [98], it is still unclear why similar results were not observed for both upper- and lower-body exercises. In that regard, future research on this topic is warranted.

4.2 Effects of Set Structure Manipulation on Velocity and Power Outputs at Submaximal Loads, Vertical Jump Performance, and Force–Velocity Profiles

Moderate GRADE quality of evidence shows that alternative set structures induce greater velocity and power outputs at submaximal loads compared to TS. However, these positive effects of alternative set structures were not substantial (SMD = 0.18). In addition, when examining the selective

effects of CS and RR, the sub-group analysis revealed that the positive effects of CS and RR over TS were not significant ($p < 0.05$). This is surprising since CS and RR improve acute energy restoration and better maintain performance than TS, and thus are thought to be beneficial for movement velocity and muscular power gains [15, 54]. While it may be tempting to conclude that CS and RR are equally effective in enhancing velocity and power as TS, it must be noted that considerably lower number of studies included in each sub-group analysis may have affected the results. This contention is supported by the pooled results of the present review, and by a recent meta-analysis [54] which compared the acute effects of TS, CS, and RR on velocity and power outputs, and concluded that both CS and RR were superior to TS, with CS providing even larger effects on acute mechanical performance enhancement. Since the present meta-analysis included a considerably lower number of studies (11 vs 19) than that on the acute effects of TS, CS and RR, it is plausible that lower statistical power caused the discrepancy between the findings. In addition, careful inspection of the two studies [55, 90] which have shown the greatest effects of CS and RR on velocity and power adaptations might explain our findings further. Interestingly, both Oliver et al. [55] and Morales-Artacho et al. [90] examined velocity and power outputs at submaximal loads that closely mirrored those used during the training intervention, whereas other studies either used less specific testing conditions, or tested against a range of specific and non-specific submaximal loads which may have skewed the results. According to the velocity specificity principle [99, 100], high-velocity training elicits greater increases in strength and power at high-movement velocity than at low-movement velocity, whereas low-velocity training develops strength and power at low-movement velocity with little effect on high-velocity strength and power, indicating the velocity-specific adaptations to RT. Although RT also causes increases in strength and power above and below the training velocity, the transfer diminishes as the testing velocity deviates from the velocity of training [101, 102]. This principle is even more relevant in this review considering that only a few included studies used a periodised approach to training while the rest of them applied little to no variation during the training intervention. Current evidence suggests [103, 104] that no or only minimal transfer effect can be expected when training specificity is offset by testing on a neutral instrument. Therefore, it may be that using a less specific condition (in this case intensity of load) to assess mechanical performance improvement during RT might also compromise the evaluation of the resultant training adaptations.

Neither the presence of muscular failure during TS, the number of repetitions performed in a set, nor the load used were significant moderators of the differences between TS and alternative set structures for velocity and power

adaptations. However, study duration was a significant moderator of the effectiveness of alternative set structures with longer duration studies promoting greater velocity and power outputs at submaximal loads compared to TS. Although interesting, this finding likely represents a type I error for the following reasons: the study [90] which reported the greatest effect of CS over TS, although lasting 11 weeks in total, manipulated set structures only in the last training block (i.e., 3 weeks), and we found no significant association between the effects of alternative set structures and the study duration when analysing RR separately (i.e., while excluding the afore-mentioned study [90]). Therefore, we suspect this analysis represents a false-positive finding given the non-significant effects of the study duration when a generally higher number of studies that compared TS and RR were analysed separately. Finally, it must be noted that, although significant, only small effects were observed in favour of alternative set structures over TS for inducing greater velocity and power adaptations. Nevertheless, even small improvements in performance in some sports may translate to meaningful differences in competitive outcomes [105, 106].

Moderate GRADE quality of evidence suggests that altering the set structure to include shorter but more frequent rest periods has marginal positive effects on vertical jump performance improvement since TS, RR, and CS were all equally effective. However, effect sizes in 4 of 5 studies, as well as examination of the confidence intervals, indicate a likely benefit in favour of alternative set structures on vertical jump performance gains. This discrepancy could likely be attributed to the lack of training specificity. Briefly, both studies that reported the smallest effects in favour of alternative set structures over TS on jumping performance did not include vertical jumps in the training program. These studies also followed a fixed (i.e., non-periodised) and highly demanding RT program. While implementing vertical jumps in a well-rounded RT program is likely to elicit the greatest improvement in vertical jump performance, it seems that training variation and specificity of load might also affect the resultant vertical jump performance adaptations [107]. This is probably also relevant to other physical capacities, such as sprinting, horizontal jumping, change of direction and throwing, since the only two studies [57, 58] which looked at these outcomes reported marginal improvements after RT regardless of the set structure. However, the results were also probably affected by the lack of training specificity since these performance-related outcomes were not a part of the training intervention. Therefore, future studies examining the effects of alternative set structures on various sport performance-related outcomes should take a periodised approach while incorporating them into a RT program to better balance stimulus and fatigue and prevent this from being a possible confounder.

Moderate GRADE quality of evidence indicates the ability of alternative set structures to modulate force–velocity profiles of the individuals. More specifically, alternative set structures induced significant shifts of the slope of force–velocity profiles towards being more velocity-oriented compared to TS. This finding is not surprising considering effect sizes for velocity and power outputs at submaximal loads and vertical jump performance gains all favoured alternative set structures and that muscular strength adaptations did not show an effect of set structure. Greater velocity and power adaptations at submaximal loads, vertical jump performance gains and consequent shifts towards a velocity-oriented force–velocity profile after utilising alternative set structures are consistent with the principle of specificity, which dictates that the more closely a training program replicates the requirements of a given outcome, the greater the transfer of the training to that outcome [108]. However, Iglesias-Soler et al. [86] raised an additional matter worthy of discussion while reporting greater shifts towards velocity-oriented profiles after RT using TS rather than RR. In that study [86], the intended velocity was always maximum, although the actual training velocity was always higher during RR than TS, which led the authors to speculate that the intended velocity matters more than the actual training velocity with respect to the changes in the force–velocity relationship. However, this training study consisted of only 10 sessions and used a unilateral leg extension exercise only with a constant load (75% of 1RM) throughout all sessions. In addition, each individual served as their own control since opposite legs performed RT using different set structures (i.e., TS or RR). While this approach minimises the biological variability that may affect training outcomes, the authors acknowledged that the cross-education effect phenomenon, although minimal and probably negligible, could have occurred in their study [86]. In that regard, the cross-education effect, together with previously mentioned factors (i.e., low number of sessions and the constant load used) may have caused similar adaptations between limbs despite differing training protocols [80]. Thus, it is presently unclear whether intended or actual velocity is the most important stimulus for shifting a force velocity profile to be more velocity-oriented. Therefore, an avenue for future research is to identify which confounding factors can affect the ability of alternative set structures to modulate force–velocity profiles.

4.3 Implications for Training and Research

Overall, the findings of this review suggest that altering the set structure during RT does not affect increases in muscle strength and size. However, the training loads and volumes were matched in the experimental groups of each study included in this review. In that regard, and considering the

mechanical advantage and lower metabolic stress and perceived exertion associated with CS and RR compared to TS, it is reasonable to assume that participants in CS and RR groups may have been able to tolerate greater training loads, perform more repetitions in a training set, or both. In fact, Iglesias-Soler et al. [109] examined the maximal number of repetitions that could be performed using RR and TS. In that study, participants performed 3 sets of squats to failure using a 4RM load using TS with 3 min of inter-set rest. During RR, participants performed multiple sets of one repetition with the same load until the point of exhaustion. In this case, rest between repetitions was calculated based on the ratio of the total recovery time during TS to the individual number of repetitions completed during TS minus one. RR allowed participants to complete about 5 times as many repetitions as the TS protocol ($RR = 45.5$; $TS = 9.3$ repetitions). Similarly, Iglesias-Soler et al. [110] showed that CS can be an effective tool for increasing training volume and intensity. In this study [110], participants performed repetitions to failure using TS and using CS consisting of 30 s of rest between every repetition. As a result, a greater number of repetitions were performed when CS was used in the bench press and biceps curl exercises, although the load for CS condition was considerably heavier (90% and 70% 1RM for CS and TS, respectively). Tufano et al. [111] expanded on these findings while showing that CS can allow for greater loads to be lifted without decreasing relative movement velocity when compared to TS, which the authors summarised as greater training stress with similar degrees of relative fatigue. Since dose–response relationships exist between RT volume and both muscle strength gains [14] and increases in muscle growth [37], these data collectively suggest that alternative set structures might be more effective at inducing these adaptations than TS if they are designed to include greater training loads and number of repetitions during RT. Indeed, Nicholson et al. [56] found that a group using a CS structure with greater loads increased their strength slightly more ($13.35 \pm 2.13\%$) than volume-equated TS ($12.16 \pm 2.76\%$) and CS ($11.14 \pm 3.35\%$) groups following a 6-week training period while still experiencing lower or similar levels of metabolic stress and decline in movement velocity. However, these potential advantages of alternative set structures are largely inferred from cross-sectional studies [110–112]. Therefore, exploring the long-term effects of CS and RR including greater loads or number of repetitions compared to TS on strength gains and muscle growth appears to be an attractive avenue for future research.

Local muscle endurance adaptations were greater after using TS rather than CS and RR structures during RT. However, the considerable amount of mechanical, metabolic and perceptual markers of fatigue often associated with TS challenges the rationale of using TS during RT to improve local muscle endurance. For instance, performing a high

volume of repetitions during RT may deteriorate movement technique due to acute fatigue [113, 114]. A strategy for achieving task outcomes in the presence of fatigue often associated with movement reorganisation involves shifting muscle recruitment away from fatigued muscles to others to compensate [115, 116]. This phenomenon, observed during the successive performance of physical tasks to fatigue [115, 116], suggests that, to maintain global task performance, the neuromuscular system shifts movement contributions away from the prime movers that fatigue earliest and towards other muscle groups still intimately involved in movement outcome. In that regard, it has been shown that weightlifting technique can be maintained to a greater extent with CS as opposed to TS [113], while the contribution from the joints responsible for producing power in the back squat exercise was altered less during RR than TS [117]. While this suggests that alternative set structures may be more effective for skill acquisition and subsequently performance enhancement, more longitudinal studies are needed to substantiate these claims. Finally, RT can be used to improve local muscular endurance via high-repetition sets at a close proximity to failure. However, this type of RT is sometimes also used to improve overall work capacity, which occurs as a result of the large number of repetitions completed [1, 2, 118]. However, work capacity can be improved more directly through aerobic training modalities, such as running [119], while also avoiding potentially maladaptive RT movement patterns [113, 117] that can occur as a result of performing a high number of repetitions within sets in a consecutive fashion. Therefore, the support for using TS during RT for the sole purpose of improving muscle endurance is tenuous.

The current findings indicate that alternative set structures are generally more effective than TS at promoting velocity and power adaptations, vertical jump performance and consequent shifts towards velocity-oriented force–velocity profiles after RT, whereas increases in muscle strength and size will not be affected by the set structure choice. In that regard, it is a viable recommendation to use alternative set structures during various training phases, such as strength–endurance, maximal strength, strength–speed, and speed–strength. For example, a strength–endurance phase often requires a high volume of repetitions to increase overall work capacity and stimulate increases in muscle cross-sectional area of an athlete [2, 118]. Since CS and RR can allow for higher volumes of work while preventing potential fatigue-induced deterioration in form, they may be preferable to TS structures. Furthermore, during a maximal strength phase, athletes often shift their training focus to exercises that emphasise force production. In that regard, alternative set structures may allow for greater loads to be lifted with lower or similar levels of metabolic stress and declines in movement velocity compared to TS, thereby training the high-force end of the force–velocity spectrum more

efficiently. Finally, both strength–speed and speed–strength phases aim to increase rate of force development while lifting relatively heavy loads as quickly as possible. Here as well, CS and RR can encourage higher velocity and power outputs, since fatigue can quickly manifest when repeatedly performing explosive movements under external load using TS [15, 54], potentially attenuating training adaptations. This is further supported by recent studies where alternative set structures allowed for a greater number of repetitions to be performed before reaching a certain velocity [120] or power loss threshold [121]. However, in some instances, the objective of RT requires a multifaceted approach to training in which hypertrophy, strength, and velocity and power outputs are all considered throughout a microcycle. This is especially prudent in team sports where frequent matches throughout the season and extended competitions periods (e.g., entering play-offs or playing Champions League) can alter the length of the preparatory period and its specific phases [122]. In that regard, alternative set structures might be a viable option for those facing time constraints since, with a proper distribution of rest during RT, it is possible to have similar (or greater) workload in training without sacrificing velocity and power outputs [56, 109–111], leading to similar hypertrophic and strength changes with greater increases in velocity and power. Nevertheless, a previous meta-analysis found that alternative set structures are highly effective at reducing acute metabolic stress and perceptual

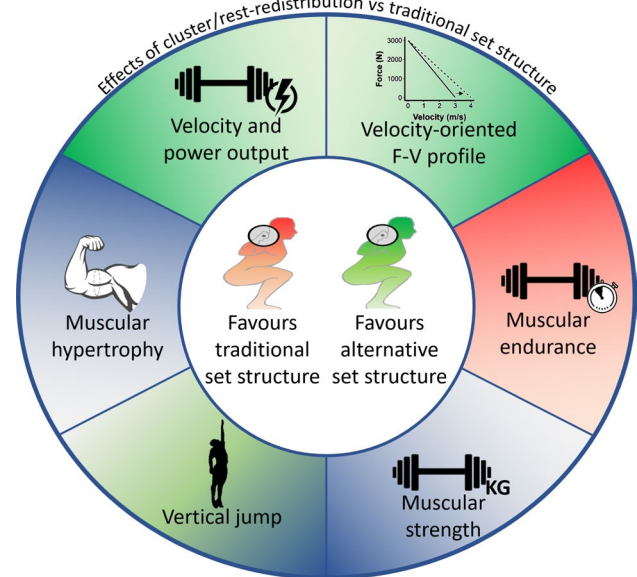


Fig. 7 Summary of chronic adaptations to resistance training using traditional and alternative (i.e., cluster and rest redistribution) set structures. Outcomes shaded in blue were not significantly different between the set structures. For the outcome shaded in green and blue (i.e., vertical jump), the results were in favour of alternative set structures compared to the traditional set structure, but these effects were not significant

exertion induced by RT when compared to TS, further suggesting their ability to accelerate the rate of recovery following RT [54]. Broadly speaking, this feature of alternative set structures would be important during in-season training when the weekly scheduling of training is dictated by the dual need to maintain (or even improve) physical qualities of the players and avoid excessive residual fatigue at the end of the week in preparation for the next match [122]. In addition, athletes who train multiple times a day and who need to manage RT fatigue for both event performance and sport-specific training sessions [21, 38, 123] might also benefit from alternative set structures. For these reasons, CS or RR structures provide coaches and athletes with a multitude of options to design their training to better manage fatigue while inducing desired adaptations. Collectively, alternative set structures can be manipulated in various ways that encourage different chronic training adaptations (Fig. 7), not just to result in less fatigue as demonstrated by previous individual studies and meta-analyses.

4.4 Limitations and Considerations

Several aspects of this review should be considered when interpreting the findings. First, analyses of hypertrophy, muscle endurance, vertical jump performance, and the slope of force–velocity profiles included a small number of studies with a combined number of participants ranging from 99 to 142. However, it should be considered that only small to moderate levels of heterogeneity were observed, suggesting that the results were broadly consistent (i.e., stable). Second, the considerably lower number of female participants in most analyses limits the generalisability of our findings to female participants, and more research should include females when possible. Third, while we performed meta-regression to examine how muscular failure, number of repetitions, study duration, and the load used during TS impacted the differences in the effectiveness between alternative set structures and TS, the number of included studies only allowed this to be performed for strength, velocity and power outcomes. In addition, while we attempted to compare the effectiveness of CS and RR set structures against each other, a relatively small (and disproportionate) number of studies prevented us from doing so. Therefore, more research is needed on the selective effects of CS and RR on various training adaptations. Fourth, some studies did not report all information required for meta-analysis and we therefore extracted the required information from figures or estimated the information (e.g., pre-post assessment correlations) based on other studies. This likely introduced some error and we therefore urge researchers to report standard deviations of the difference scores (and or pre-post assessment correlations) when reporting the results of training intervention studies. Similarly, only four studies reported

adherence with the intervention, and future studies should therefore ensure that adherence rates for all groups are presented. Fifth, almost half of the studies included in this review were considered to be at high risk of set structure bias since the participants were performing other standardised RT sessions or because they were not instructed to perform all exercises during an intervention using a single-set structure, but rather a combination of set structures (e.g., CS or RR for the back squat, TS for the accessory exercises). While this likely increased the ecological validity of the findings, it also meant that findings could not be solely attributed to the true effects of set structure manipulation. Finally, some studies did not include specific exercises or intensities during the training, despite being assessed at the end of the intervention (e.g., vertical jumps were not included in training, but their improvement was assessed). This could have caused an underestimation of the actual differences between TS and alternative set structures for certain outcomes.

5 Conclusion

This systematic review and meta-analysis found that set structure manipulation during RT has a negligible impact on muscular strength gains and hypertrophic outcomes. However, it should be noted that training intensities and volumes were equated between the set structures in all included studies. Since alternative set structures could allow for greater training loads and volumes when compared to TS, future studies should explore the long-term effects of CS and RR including greater loads or number of repetitions compared to TS on strength gains and muscle growth. While using CS and RR during RT generally led to greater vertical jump performance when compared to training using TS, the difference between performance gains was not significant. This was likely caused by the training specificity issues observed in many studies since vertical jump was rarely a part of the RT protocols. Alternative set structures induced greater velocity and power outputs at submaximal loads than TS and were also able to modulate force–velocity profiles towards being more velocity-oriented. TS had more favourable effects on local muscle endurance improvement when compared to CS and RR. However, since a considerable number of mechanical, metabolic and perceptual markers of fatigue are often observed during TS, and because there are other, more effective ways of improving work capacity, the larger improvement in muscular endurance during TS compared to CS and RR may be of limited practical relevance. Collectively, the findings of this review indicate that alteration of TS to alternative set structure (CS or RR) is expected to influence the magnitude of specific muscular adaptations which further suggests set structure manipulation is an important consideration for RT program design.

Compliance with Ethical Standards

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Ethics approval Not applicable.

Availability of data and material The datasets generated during and/or analysed during the current review are available from the corresponding author on reasonable request.

Consent Not applicable.

Author contributions IJ performed the meta-analysis and meta-regression and wrote the first draft of the manuscript. All authors edited and revised the manuscript and approved the final version of the manuscript.

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